

Data Science 6

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Lecture 7

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Visualizations

0.1 Examples

We finna learn about matplotlib and seaborn. Visualizing data helps us better understand our data. Recall the cholera water pump example. Rose diagram Florence Nightangale.

0.2 What is visualization?

How we visaulize narrates the story and allows us to create a "purpose/meaning" for the reader.

Common Practices:

- Make easy to read
- Accurate
- Use good titles and labels

0.2.1 Line Plot

Good for visualizing 2 num vars like x and y. One y value for each x value. X-values should have a clear sequential order.

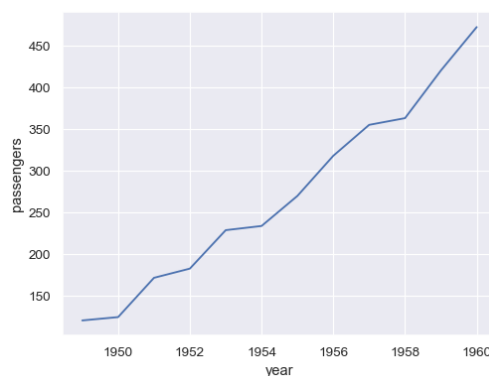


Figure 1: Example line plot

0.2.2 Scatter Plot

Looking for corelations/assocaitons and have no sequential data. There are some other cool visuliaztions as well

0.2.3 Bar Chart

Good for categorical data. Ex: spotify wrapped

1 Code

Use **.scatter** (scatter plot) or **.plot** (for line plot) to create tables we saw before. We looked at num movies and total gross income realized line plot was not good since x was not sequential and looked bad. Looking for association instead so we used a scatter plot. The **.group** works by counting by category and then we can use **.barh** to create a bar chart to rep the counts for each category.

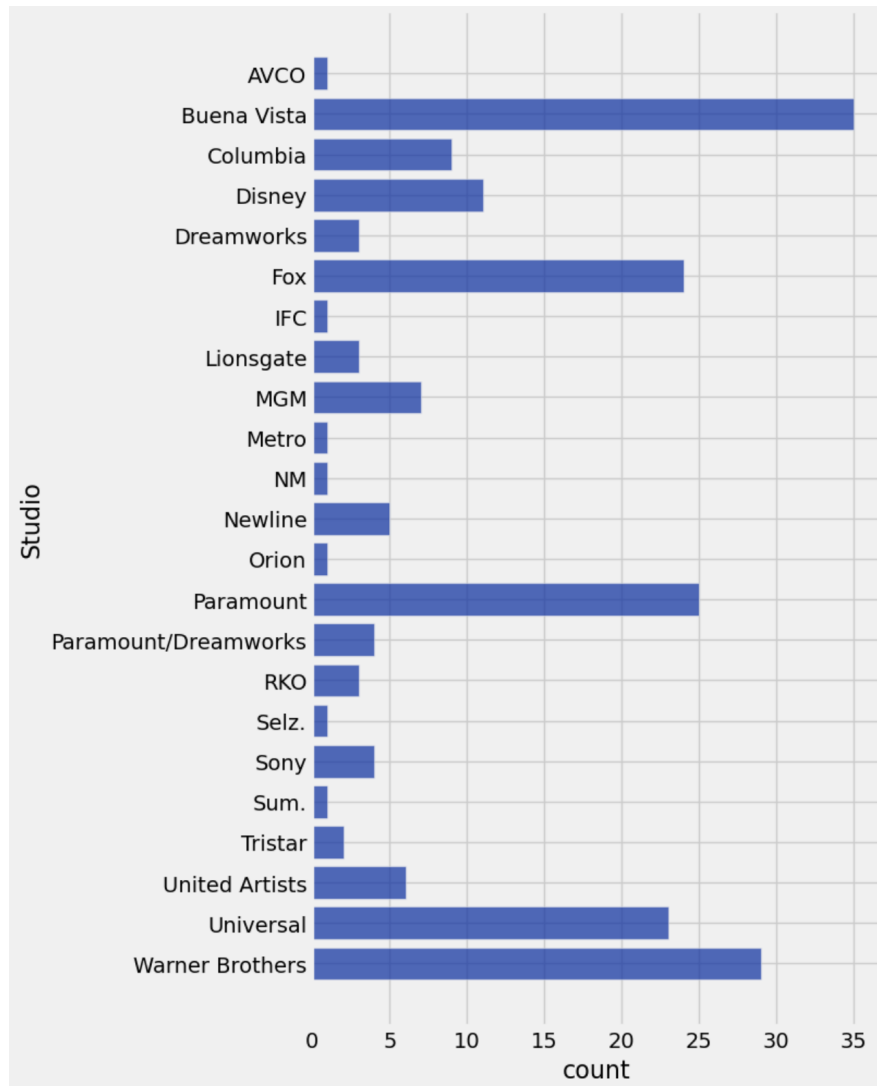


Figure 2: Bar chart.